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PMODefTopic



About the PMO

About the PMO

The Precision Meter Option (PMO) is a high accuracy digital multimeter option for the UltraFLEX test system that includes an Agilent 3458A digital multimeter mounted in the options space in the support cabinet. The PMO connects to the General Purpose Interface Option (GPIO) in the test head and can be used to:

- [Make high accuracy voltage, current, or resistance measurements](#)
- [Calibrate references on a DIB](#)
- [Calibrate the DIB](#)
- [Perform NIST traceability](#)
- [Make DUT measurements](#)

PMO measurements are performed by the Agilent 3458A meter in its low-level [Agilent Multimeter Language](#), which is documented in the [Agilent 3458A Multimeter Operating, Programming, and Configuration Manual](#) delivered with your tester. To free you from the time-consuming details of HPML programming, Teradyne has developed a high-level PMO language on top of the HPML language to perform commonly-used measurements, such as:

- [Voltage measurements](#)
- [Current measurements](#)

- [Voltage ratio measurements](#)
- [Two-wire resistance measurements](#)
- [Four-wire resistance measurements](#)

If you prefer the flexibility of the HPML, the PMO subsystem gives you direct access to the meter through a General Purpose Interface Bus (GPIB). The GPIB allows you to operate the meter using HPML commands. Like the meter itself, the GPIB has its own programming language, known as the GPIB language—for controlling traffic on the bus. Using GPIB programming, you can send GPIB commands to the meter and receive control and status information from the meter. The PMO also allows you to combine PMO and GPIB statements.

PMO information includes the following sections:

- [PMO Theory of Operation](#)
- [PMO Programming](#)
- [PMO Debug Display](#)
- [PMO Examples](#)

Related Topics

- [Precision Multimeter Option Specifications](#)
- [PMO \(VBT syntax\)](#)
- [GPIB Programming](#)



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About the PMO : PMO Software Licensing Requirements

PMO Software Licensing Requirements

Operation of the PMO does not require a software license.

For more information on IG-XL licensing, see [IG-XL Licensing](#) in IG-XL Administration help.

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PMO Debug Display

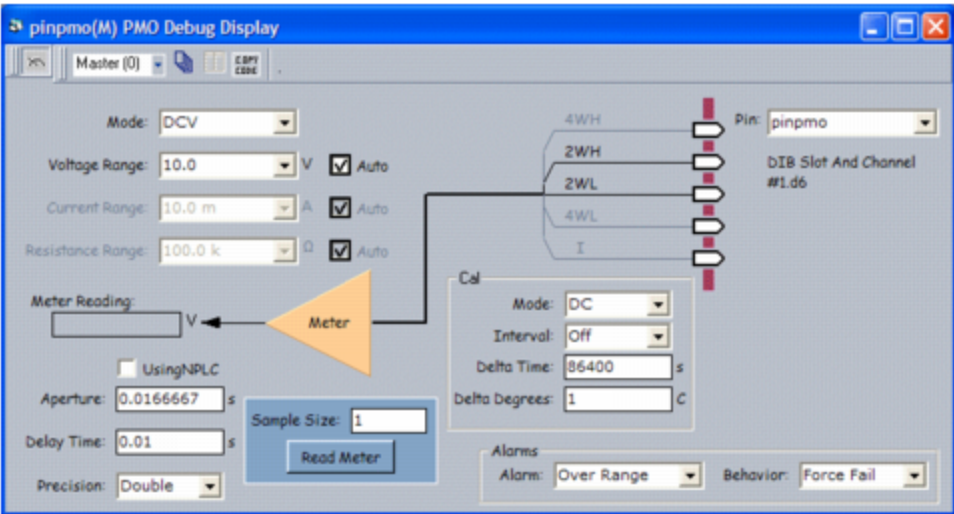
PMO Debug Display

IG-XL includes a debug display for the PMO. This display is an interactive, window-based tool which assists in the development of test programs that use the PMO. Use the display to determine the state of the PMO hardware and to interactively set PMO parameters.

If used offline, the PMO debug display displays the cached software state of the meter.

This section describes features specific to the PMO debug display. For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays



PMO Debug Display

To use the PMO Debug Display you must have a test program loaded and validated. Bring up the display as follows:

1. [Start the Test Debug Environment by clicking the](#) Test Debug Environment button in the One-Click Launch group on the IG-XL Tools tab.
2. [Select](#) Instruments>PMO.

You can also bring up the PMO debug display using the VBT statement:

```
TheExec.Tools("PMO").ShowPin (" <PinName> ")
```

The PMO display includes the following controls:

Mode	Determines the enabling and disabling of the range combo boxes and their Auto range checkboxes. For example, if you choose DCV as the Mode, then only Voltage Range combo box and its Auto range checkbox are enabled. The Current Range and Resistance Range combo boxes and their corresponding Auto range checkboxes are disabled.
Voltage Range	Displays the programmed voltage range.
Auto (Voltage Range)	Applies the auto range to the voltage range.
Current Range	Displays the programmed current range.
Auto (Current Range)	Applies the auto range to the current range.
Resistance Range	Displays the programmed resistance range.
Auto (Resistance Range)	Applies the auto range to the resistance range.
Meter Reading	Displays the PMO reading. If the sample size is 1, the meter displays that reading. If the sample size is greater than 1, the meter displays the average of the meter readings.
Using NPLC	Checking this box selects NPLC. Clearing this box selects Aperture. This text box allows you to program the measurement integration time of the meter. You can enter an integer in this text box to program the measurement integration time of the meter. If you pick the Aperture option after programming Using NPLC, the Aperture field shows the actual aperture time corresponding to the NPLC that has been programmed to the meter. Using NPLC is for programming only. If you pick the NPLC option again, Using NPLC comes up empty. It will be filled with the number that you entered the next time you program the NPLC

but will be cleared whenever you switch from the Aperture option to the NPLC option.

Aperture

Displays the aperture time over which the meter's A/D converter measures the input signal. Longer aperture or integration times average out noise and increase the resolution of the digital output from the converter, thereby increasing measurement accuracy and decreasing measurement speed. Aperture and NPLC (the number of power line cycles) both set the integration time. Executing either cancels the integration time previously established by the other. The PMM integration time reset state is $NPLC = 1$.

You can enter new or different Aperture values to program the measurement integration time of the meter. This is also the initialization state of the PMO debug display.

Delay Time

Displays the internal delay time for PMO software. This is the time in seconds that IG-XL waits before strobing and reading the PMO meter.

Precision

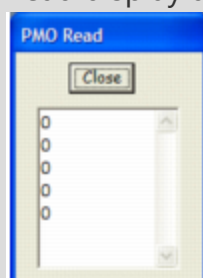
Displays the precision of the PMO meter reading. Can be Single or Double.

If single precision is displayed, IG-XL returns a single precision real number into a variable of type double. Single precision is equivalent to the float data type, which represents real numbers with up to 5.5 digits of precision. Single precision numbers are four bytes long. It takes IG-XL half the time to transfer a single precision meter reading, as compared to a double precision reading. The meter also uses half as much memory to store single precision numbers.

If double precision is displayed, IG-XL returns a double precision real number into a variable of type double. Double precision numbers are eight bytes long. Double precision must be used when greater than 5.5 digits of precision are required.

Sample Size

The number of samples to be taken for the measurement. If you enter a number greater than 1 and click the Read Meter button, a separate PMO Read display appears.



Each read result is listed in the PMO Read display with the first reading on top, the second below, and so on. Use the scrollbar to see read results that are outside the viewing area. The average of the values in the PMO Read display appears in the Meter Reading field. To close the PMO Read display, click the Close button.

Read Meter

Displays the meter reading. If the Sample Size is greater than 1, this button brings up the PMO Read display. See [Sample Size](#).

4WH, 2WH, 2WL, 4WL, I

The 2-wire and 4-wire connections from the PMO to the Pogo pins. Selecting the mode displays the appropriate connections. For example, if the meter is set to DCV mode, the 2WH and 2WL lines are visible while the other three lines are grayed out.

Cal Mode

Displays which calibration routine or routines are selected. When to run them is controlled by the Cal Interval and Cal Delta Time options.

- [DC](#): The DC calibration routine has been selected. This routine maintains the accuracy of all measurements
- [AC](#): The AC calibration routine.
- [Ohms](#): The Ohms calibration routine.
- [All](#): All PMO calibration routines: AC, DC, and Ohms.
- [<mode list>](#): Is a combination of calibration routines, enclosed in parentheses and separated by spaces. For example, ac ohms indicates the AC and Ohms calibration routines.

Cal Interval

Specify how often to calibrate.

Cal Delta Time

Specify the interval (elapsed time) in seconds that causes calibration.

Cal Delta Degrees

Specify the change in temperature in degrees Celsius that causes calibration.

Alarms: Over Range

Indicates an overrange condition. If this alarm occurs, the number 1E+38 is returned.

Alarms: Behavior

- **Force Fail:** Forces test to fail if alarm occurs. If no alarm is specified, this parameter forces fail on all alarms.
- **Force Bin:** Forces the test to bin the device if the specified alarm occurs. If no alarm is specified, this parameter forces binning on all alarms.
- **Off:** Ignores the specified alarm. If no alarm is specified, this parameter ignores all alarms.
- **Default:** Restores behavior to Force Fail.
- **Continue:** Reports the alarm, but does not change the DUT status.
- **Raise Error:** Reports the alarm by returning an error.

Pin

The menu allows you to select pinnames. Below the pinname is the associated DIB channel string. Any pin in the pin map that has a PMO behind it is a valid choice. This channel string is the DIB pin (DibSlotPin) where the PMO is connected on the DIB.

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PMO Examples

PMO Examples

Set Up DataTool

The following DataTool sheets and data are used to program the PMO:

- Pin Map
- Channel Map
- Test Procedures
- Test Instances
- Flow Table

PMOExample.xls					
	A	B	C	D	E
1	Pin Map				
2					
3	Group Name	Pin Name	Type	Comment	
4		pinPMO	Analog		
5					

Pin Map Sheet

PMOExample.xls						
	A	B	C	D	E	F
1	Channel Map					
2						
3	DIB ID:		View Mode: Pogo			
4						
5	Device Under Test		Tester Channel			
6	Pin Name	Package Pin	Type	Site 0	Comment	
7	pinPMO		PMO	1.06		
8						

Channel Map Sheet, Pogo View

PMOExample.xls						
	A	B	C	D	E	F
1	Channel Map					
2						
3	DIB ID:			View Mode: Signal		
4						
5	Device Under Test			Tester Channel		
6	Pin Name	Package Pin	Type	Site 0	Comment	
7	pinPMO		PMO	1.pmo		
8						

Channel Map Sheet, Signal View

PMOExample.xls				
	A	B	C	BN
1	Test Procedures			
2				
3				
4	Procedure Name	Element	Nop	Comment
5	PDEExample	Variable		
6	PDEExample	Variable		
7	PDEExample	Variable		
8	PDEExample	Variable		
9	PDEExample	LevelsTiming	TRUE	
10	PDEExample	PMOSetup		
11	PDEExample	PMOMetering		
12	PDEExample	Limits		
13	VBTEExample	Variable		
14	VBTEExample	Variable		
15	VBTEExample	Variable		
16	VBTEExample	LevelsTiming	TRUE	
17	VBTEExample	VT		
18	VBTEExample	Limits		

Test Procedures Sheet

PMOExample.xls						
	A	B	C	D	E	F
1	Test Instances					
2						
3						
4	Test Name	Type	Name	Called As	Category	Selector
5	mytestVBT	Test Procedure	VBTEExample			
6	mytestTDE	Test Procedure	PDEExample			

Test Instances Sheet

PMOExample.xls								
	A	B	C	D	E	F	G	H
1	Flow Table							
2								
3								
4	Label	Enable	Job	Part	Env	Opcode	Parameter	TName
5						Test	testVBT	
6						Test	mytestTDE	
7						stop		
8								

Flow Table

For more information, see:

- [PMO \(VBT Syntax Reference\)](#)

	• PMO Programming
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PMO Programming

PMO Programming

This section describes programming of the PMO.
The following topics are available:

- [Programming the Configuration File and Cable File](#)
- [PMO DataTool Programming](#)
- [PMO Reset Values](#)

For more information, see:

- [PMO \(VBT Syntax Reference\)](#)
- [PMO Examples](#)

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PMO Programming : Programming the Configuration File and Cable File

Programming the Configuration File and Cable File

To use a PMO, a test system must have the these files:

- [One or more Simulated Configuration Files](#) with an entry in the GPIB section
- [A PMO Cable File](#) that describes the PMO connections

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PMO Programming : Programming the Configuration File and Cable File : Simulated Configuration Files

Simulated Configuration Files

The test system configuration file lists the cards in each test head slot and their connections. You must create a test system configuration file for your test system with the PMO entry. Do this by creating a SimulatedConfig.txt file and placing the file in any of the following locations:

- Folder containing the IG-XL workbook
- \tester folder: C:\Program Files (x86)Teradyne\IG-XL\<version>\tester
- \bin folder: C:\Program Files (x86)\Teradyne\IG-XL\<version>\bin

For example, for online mode, the test system configuration file could be in:
C:\Program Files (x86)\Teradyne\IG-XL\<version>\tester\TesterConfig.txt

For offline mode, the test system configuration file could be in:
C:\Program Files (x86)\Teradyne\IG-XL\<version>\tester\SimulatedConfig.txt

The PMO entry in the configuration file has the following form:

```
<GPIB CNIGPIBController0 0>
#Addr[.SndAddr] LName PName  RT  WT EOI Timeout DIBSlot  CableFile
22          PMO  HP3458A CRLF CR T  10    1      PMOCableAddr22.txt
</GPIB CNIGPIBController0 0>
```

where:

Addr[.SndAddr]	Is for PrimaryAddress[.SecondaryAddress]. PMO has only a primary GPIB address; its secondary GPIB address defaults to 0. The PMO has address 22.
LName	Is the logical instrument name for the Teradyne supported GPIB instrument. For the PMO, the LName is PMO.
PName	Is the physical instrument name for the Teradyne supported GPIB instrument. The PMO is the Agilent 3458A meter: For the PMO, the PName is HP3458A.
RT, WT, EOI	Are read terminator, write terminator, and EOI line assertion respectively. They define how GPIB data transfers are terminated for the PMO. The Teradyne supported Agilent 3458A meter uses: • CRLF (carriage return line feed) as read terminator • CR (carriage return) as write terminator • T (True) for EOI which means assert EOI at the end of a transfer.
Timeout	Is the timeout value for each GPIB data transfer in seconds.
DIBSlot	Specifies the DIB slot that the GPIB instrument is connected to. The standard PMO is one Agilent 3458A meter connected to DIB slot 1 through the GPIO.
CableFile	Specifies the channel and signal information for the GPIB instrument. The PMO has a cable file name. PMOCableAddr22.txt and the file is located in the \bin folder of the installed version of IG-XL. See PMO Cable File .

For more information, see

- [Example Test System Configuration File in Tester](#)
- [Example Configuration File for Simulator](#)
- [Tester Configuration](#)

Example Test System Configuration File in Tester

...

<Version>

1

</Version>

...

<TEST_HEAD TH 1>

#slot[.subslot] Type idprom (type serial rev company)

1.0 BBAC-15 805-004-02 0025EE92 0337-D 5445

...

DC-30 949-901-01 06000C1F 0316-B 5445

aka: 949-901-00

</TEST_HEAD TH 1>

<GPIB CNIGPIBController0 0>

#Addr[.SndAddr] LName PName RT WT EOI Timeout DIBSlot CableFile

22 PMO HP3458A CRLF CR T 10 1 PMOCableAddr22.txt

</GPIB CNIGPIBController0 0>

<LICENSE_ENABLERS>

...

Example Configuration File for Simulator

Example Test System Configuration File for Simulator

...

<Version>

1

</Version>

<TEST_HEAD TH 1>

#slot[.subslot] Type idprom (type serial rev company)

-1 dib 239-001-00 00000000 0000-A 0000

...

</TEST_HEAD TH 1>

<GPIB CNIGPIBController0 0>

#Addr[.SndAddr] LName PName RT WT EOI Timeout DIBSlot CableFile
22 PMO HP3458A CRLF CR T 10 1 PMOCableAddr22.txt
</GPIB CNIGPIBController0 0>

If you copy this file to the tester directory to specify an optional cable,
a corresponding cable file MUST EXIST in the directory "tester\cables"
(i.e., "tester\cables\866-260-23.txt")

...

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PMO Programming : Programming the Configuration File and Cable File : PMO Cable File

PMO Cable File

The PMO cable file specifies the valid Pogo entry and signal entry for the PMO. Although one PMO has eight Pogos and signals, they all come from the same meter and cannot be controlled by IG-XL independently. Thus, you can have only one valid Pogo entry and one valid signal entry for one PMO. The two-wire low Pogo D6 is used as the valid Pogo entry for the Teradyne supported PMO in a channel map because it is used in every measurement of the PMO. The PMO cable file must also have information for all DIB Pogos and signals for the PMO associated with this cable file, but they are entered as comments for your reference and cannot be used as entries in a channel map. IG-XL uses the Pogo and signal entry as part of the information to construct channel information for PMO.

See Example PMO Cable File.

Example PMO Cable File

UltraFLEX test systems are supplied with the required PMO cable file, PMOCableAddr22.txt, in C:\Program Files (x86)\Teradyne\IG-XL\NN.nn\bin.

You need not create a PMO cable file.

The contents of the PMO cable file is similar to the following:

```
# PMO
```

```
# This is a DIB Pogo and Signal map for the HP3458A PMO supported by Teradyne.
```

```
# Valid entry for this PMO in channel map
```

```
# DIB Pogo  Signal Name
```

```
D6          PMO
```

```
# Following lists all DIB Pogo and actual signals for this PMO but they are
```

```
# for reference only and are not valid for channel map entries.
```

A7 4WH
B7 4WL
B6 2WH
D6 2WL
A6 2WHs
C6 2WLs
C7 I
D7 Is

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PMO Programming : PMO DataTool Programming

PMO DataTool Programming

This section describes how to program several DataTool worksheets to handle basic PMO parameters. It includes the following topics:

- [Programming the Pin Map](#)
- [Programming the Channel Map](#)

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PMO Programming : [PMO DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the device pins by assigning a pin type to each device pin name. Pins associated with the PMO must have Analog or Unknown as the type on a Pin Map sheet. For more information, see [Pin Map Sheet](#) in the DataTool section.

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PMO Programming : PMO DataTool Programming : Programming the Channel Map

Programming the Channel Map

A Channel Map sheet assigns individual device pin names to test system channels and power supplies. You can enter pins associated with the PMO on a Channel Map sheet with Pogo view mode or Signal view mode.

For both modes, the tester channel type is PMO.

In Pogo view mode, the channel entry for a site is:

DIBSlot.DIBChannel

In Signal view mode, the PMO channel entry for a site is:

DIBSlot.Signal

Only one DIBChannel and one Signal name can be used as valid entries in a channel map for one PMO meter. The valid DIB channel/Pogo and the valid signal for the PMO are defined in its cable file described in Programming the Configuration File and Cable File.

Only a direct connection is allowed for the PMO. Pin sharing and site sharing are supported by the PMO.

For more information, see Channel Map Sheet in the DataTool section.

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PMO Programming : PMO Reset Values

PMO Reset Values

In the PMO reset state, the meter is set to DC Voltage measurements with:

- Auto Range on
- Calibration disabled
- Calibration mode set to DC
- Precision set to double
- Current Range 10 mA
- Voltage range 10 V
- Resistance Range 100 Kohms
- Delay time 10 ms
- Aperture time set to a value that corresponds to integration time of one power line cycle (one power line cycle, which is 16.67 ms for 60 Hz power and 20 ms for 50 Hz power)

Powering on a tester resets the PMO. You can also reset the PMO at any time from within a test program using the PMO VBT reset statement (see [Reset](#) in the VBT Syntax Reference).

The command in the reset statement that's sent to the PMO is:

"PRESET NORM;T SYN;AZERO OFF;MFORMAT DREAL;ERR?;AUXERR?;DISP OFF;END ALWAYS"

where PRESET NORM is an HPML command that resets the meter to its standard reset state.

The reset also clears the ERR register and the AUXERR register.

The other HPML commands in the reset_code statement adjust some of the normal state settings to make the reset state of the HP meter satisfy the needs of the PMO.

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PMO Theory of Operation

PMO Theory of Operation

This section describes the operation of the Precision Meter Option (PMO). The following topics are available:

- | | |
|---|--------------------------------------|
| • | PMO Connection |
| • | PMO Calibration |
| • | Types of Measurement |

For information on PMO instrument performance specifications and standards, see [Precision Multimeter Option Specifications](#).

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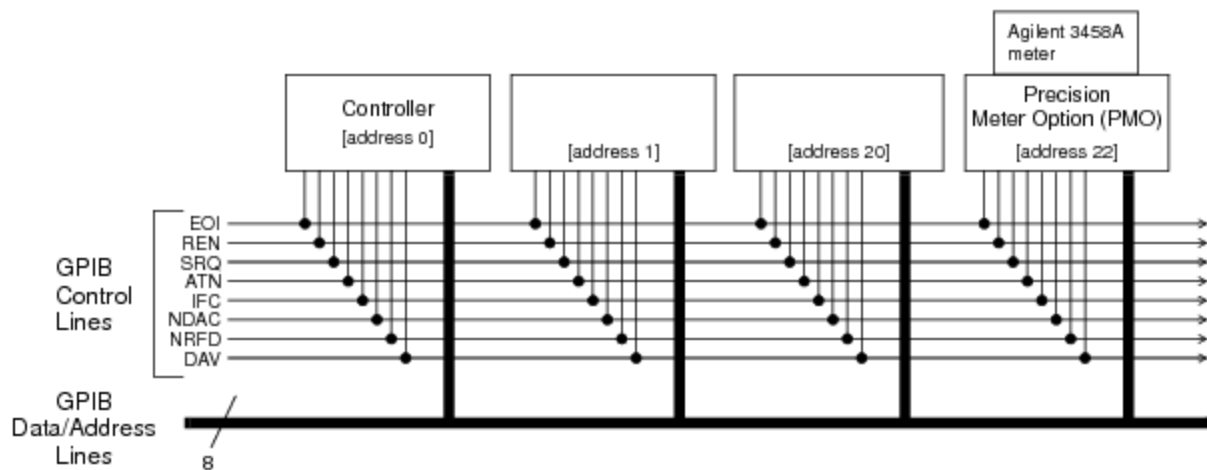
pmo_too.2.02



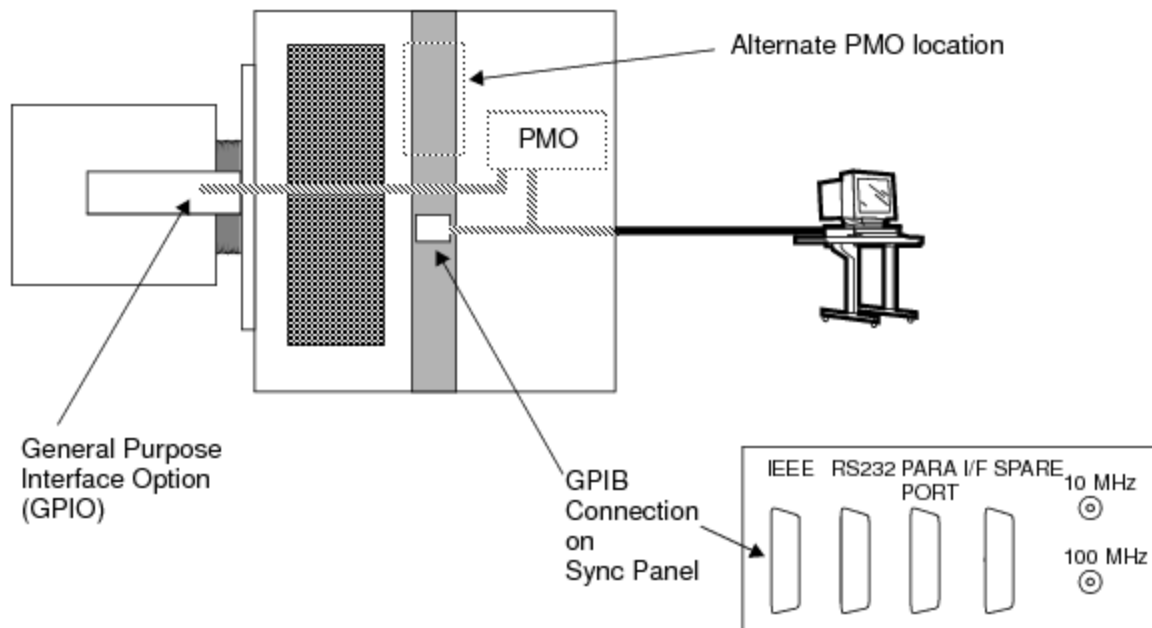
PMO Theory of Operation : PMO Connection

PMO Connection

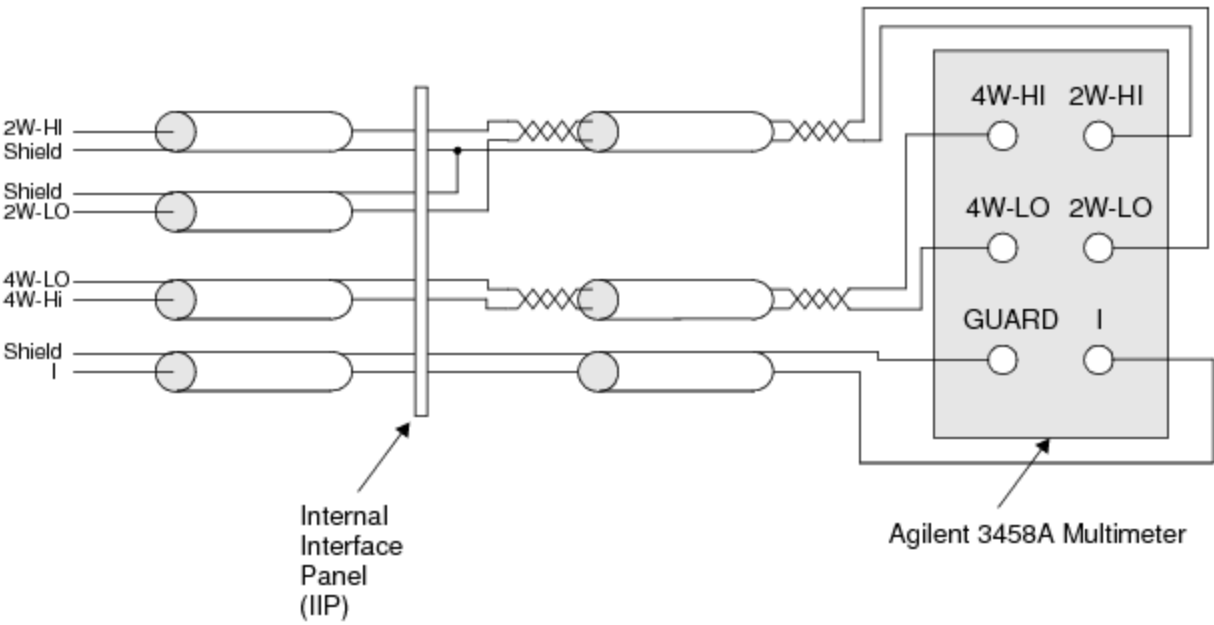
The PMO is mounted in the test system cabinet and is connected to the GPIO in the test head. The PMO GPIB address is 22.



PMO GPIB Connection



PMO Connection



PMO Cable Connections

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PMO Theory of Operation : PMO Calibration

PMO Calibration

To ensure that the PMO meets its rated accuracy specifications, you must calibrate it regularly. The PMO has two types of calibration: external calibration and internal calibration. The following calibration topics are available:

- External Calibration
- Internal Calibration
- Accuracy Dependence on Operating Temperature

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[PMO Theory of Operation](#) : [PMO Calibration](#) : External Calibration

External Calibration

External calibration involves calibrating the PMO’s internal 10V and 10kohm reference standards against external standards traceable to NIST (National Institute of Standards and Technology) or another country’s national standards. See the UltraFLEX Service Manual for the PMO’s external calibration procedure. Perform external calibration every 90 days. Teradyne externally calibrates every meter before shipping it.

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PMO Theory of Operation : PMO Calibration : Internal Calibration

Internal Calibration

Internal calibration involves calibrating the PMO’s working references against its internal 10V and 10 kohm reference standards. The PMO has three calibration routines:

DCV	This routine makes general adjustments to all the meter’s measurement functions.
ACV	This routine makes further calibration adjustments to the aci, acv, acdcv, and acdci functions.
OHMS	This routine makes even further calibration adjustments to the aci, dci, ohms_2w and ohms_4w functions.

While a calibration routine is running, you cannot use the meter nor can you execute IG-XL functions.

The execution times for the calibration routines are:

- 2.2 minutes for DCV
- 1.1 minute for ACV
- 11 minutes for OHMS
- 12 minutes for all routines, when run with cal_mode set to all

 **Note:**

Calibration times for a 50 Hz test system are 16% more than the values listed above.
To ensure that PMO measurements meet their specified accuracies:

- Run all relevant calibration routines once every 24 hours.
- Calibrate the meter only after it has been properly warmed up, that is, the tester and meter have been on for at least one hour with the cabinet doors closed.

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PMO Theory of Operation : PMO Calibration : Accuracy Dependence on Operating Temperature

Accuracy Dependence on Operating Temperature

The PMO is most accurate when its operation temperature is within 1C of the temperature at which it was calibrated (TACAL) and within 5C of the temperature at which it was externally calibrated (TXCAL). The PMO’s basic specifications only apply to measurements made within these temperature ranges. The meter keeps a record of TACAL and TXCAL, which is available on request. Because it may be difficult to maintain the meter’s operating temperature within such a narrow range, Teradyne provides a set of temperature coefficients for establishing the PMO’s accuracy outside this temperature range. The units for these temperature coefficients are error (PPM or %) per degree centigrade. For a list of these temperature coefficients, see Precision Multimeter Option Specifications.

The temperature coefficients listed for acv and aci mode are for operating temperatures greater than 1C of TACAL but within 5C of TACAL. (TXCAL is irrelevant.) To establish the new PMO accuracy specifications for this temperature range:

1. Calculate the difference between the meter’s current operating temperature and TACAL.
2. Multiply by the appropriate temperature coefficient.
3. Add the result to the PMO’s basic accuracy specifications.

Operation beyond 5C of TACAL is not specified.

The modes dcv, dci, ohms_2w, and ohms_4w have two sets of temperature coefficients identified as with ACAL and without ACAL. The with ACAL temperature coefficients are for operating temperatures within 1C of TACAL but more than 5C outside of TXCAL. To establish the new PMO accuracy specifications for this temperature range:

- | | |
|----|--|
| 1. | Calculate the absolute difference between TXCAL and the meter’s current operating temperature. |
| 2. | Subtract 5C. |
| 3. | Multiply by the appropriate temperature coefficient. |
| 4. | Add the result to the PMO’s basic accuracy specifications. |

The without ACAL temperature coefficients are for operating temperatures that are not within 1C of TACAL and not within 1C of TXCAL. To establish the new PMO specifications for this temperature range:

- | | |
|----|--|
| 1. | Calculate the absolute difference between TXCAL and the meter’s current operating temperature. |
| 2. | Multiply by the appropriate temperature coefficient. |
| 3. | Add the result to the PMO’s basic accuracy specifications. |

This mode is the least accurate.

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This section describes how to program the PMO to perform the following types of measurements:

- | | |
|---|-----------------------------------|
| • | Voltage Measurements |
| • | Current Measurements |
| • | Voltage Ratio Measurement |
| • | Two-Wire Resistance Measurements |
| • | Four-Wire Resistance Measurements |

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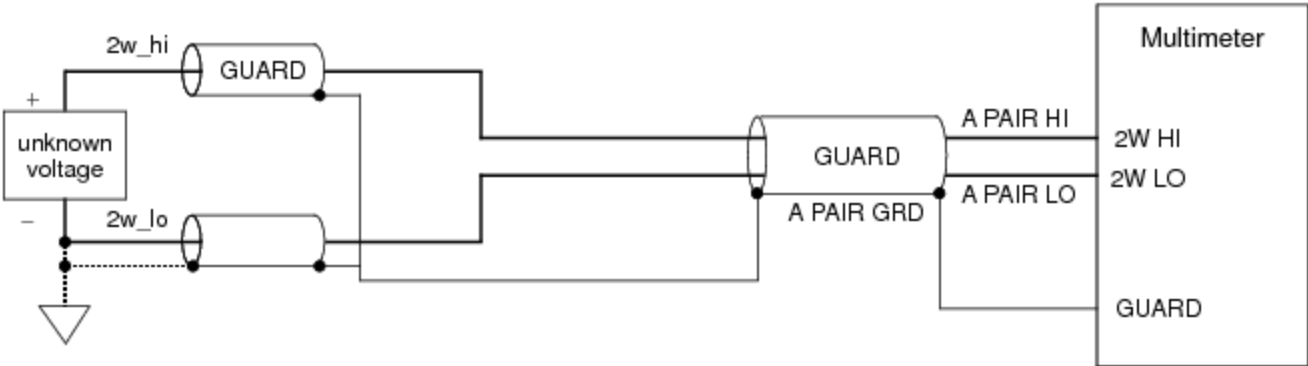
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PMO Theory of Operation : Types of Measurement : Voltage Measurements

Voltage Measurements

The PMO performs dc, ac, or dc+ac voltage measurements, using a two-wire configuration. In the two-wire configuration, two channels serve as the HI and LO inputs, which allows you to perform voltage measurements.

The two-wire configuration for voltage measurements using channel A is shown in the following figure.



Two-Wire Configuration for Voltage Measurement

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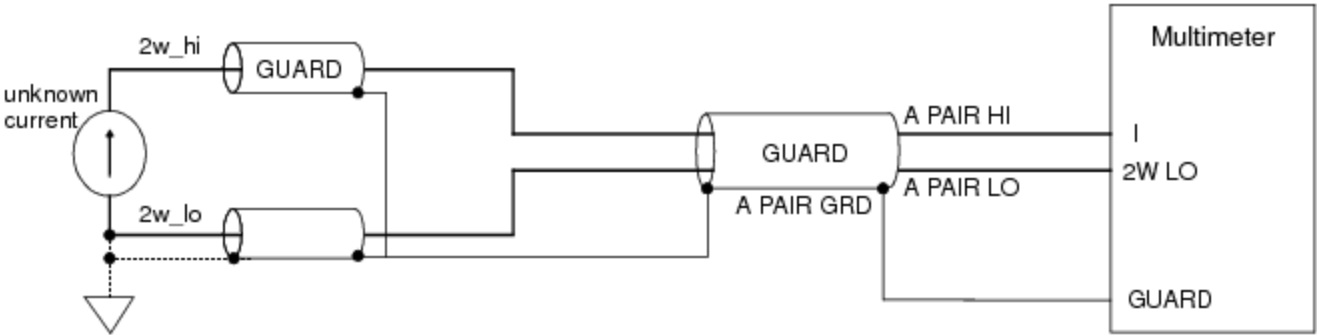
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PMO Theory of Operation : [Types of Measurement](#) : Current Measurements

Current Measurements

As with voltage measurements, PMO performs dc, ac, or dc+ac current measurements, using a two-wire configuration. In the two-wire configuration, two channels serve as the HI and LO inputs, which allows you to perform current measurements. The two-wire configuration for current measurements is shown in the following figure.



Two-Wire Current Measurement

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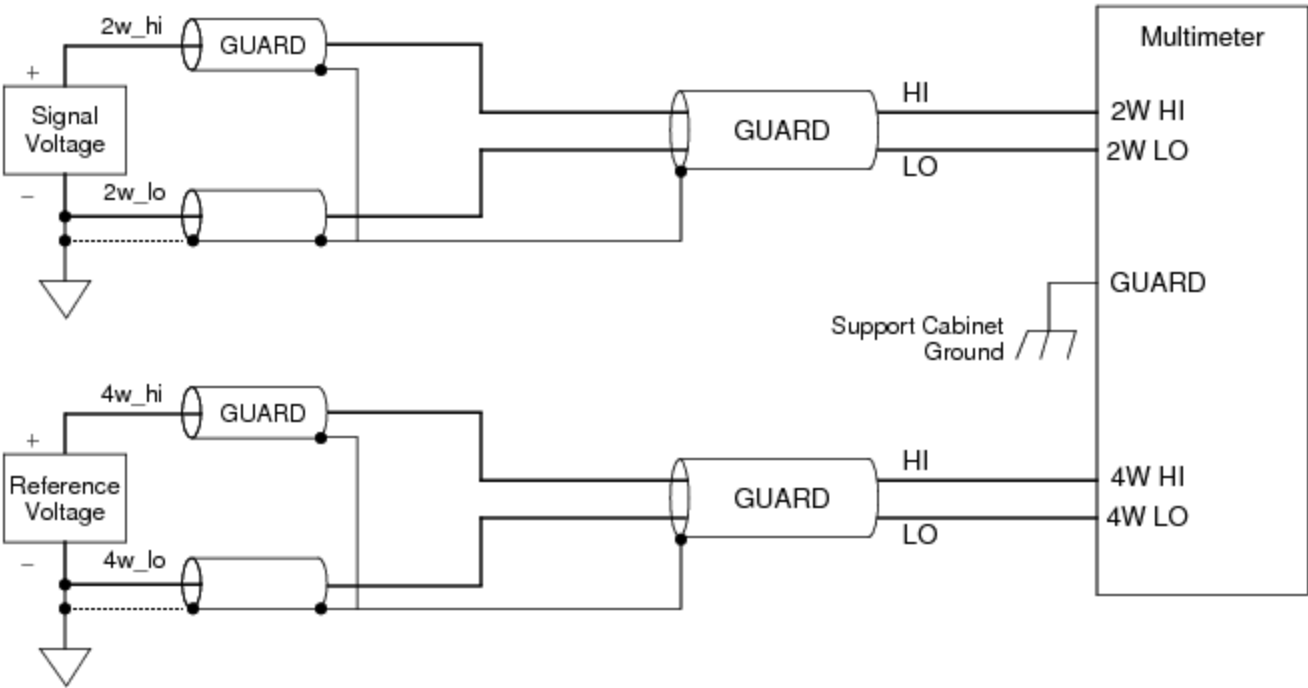
PMO Theory of Operation : Types of Measurement : Voltage Ratio Measurement

Voltage Ratio Measurement

For voltage ratio measurements, the PMO measures a dc reference voltage and a signal voltage then computes their ratio:

Voltage ratio = Signal voltage / DC reference voltage

The configuration for a voltage ratio measurement is shown in the following figure.



Voltage Ratio Measurement

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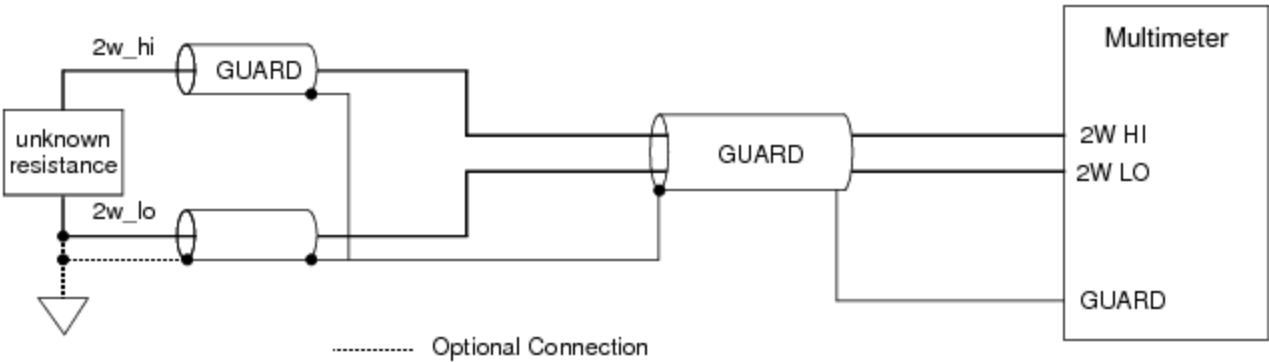
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PMO Theory of Operation : Types of Measurement : Two-Wire Resistance Measurements

Two-Wire Resistance Measurements

Use the PMO's two-wire configuration for measuring resistances greater than 1 Mohm. For resistances below 1 Mohm, the loop resistance between the DUT and the meter reduces measurement accuracy. In this case, use the four-wire configuration to lessen the effect of the internal loop resistance (see Four-Wire Resistance Measurements).

The two-wire configuration for resistance measurements is shown in the following figure.



Two-Wire Resistance Measurement

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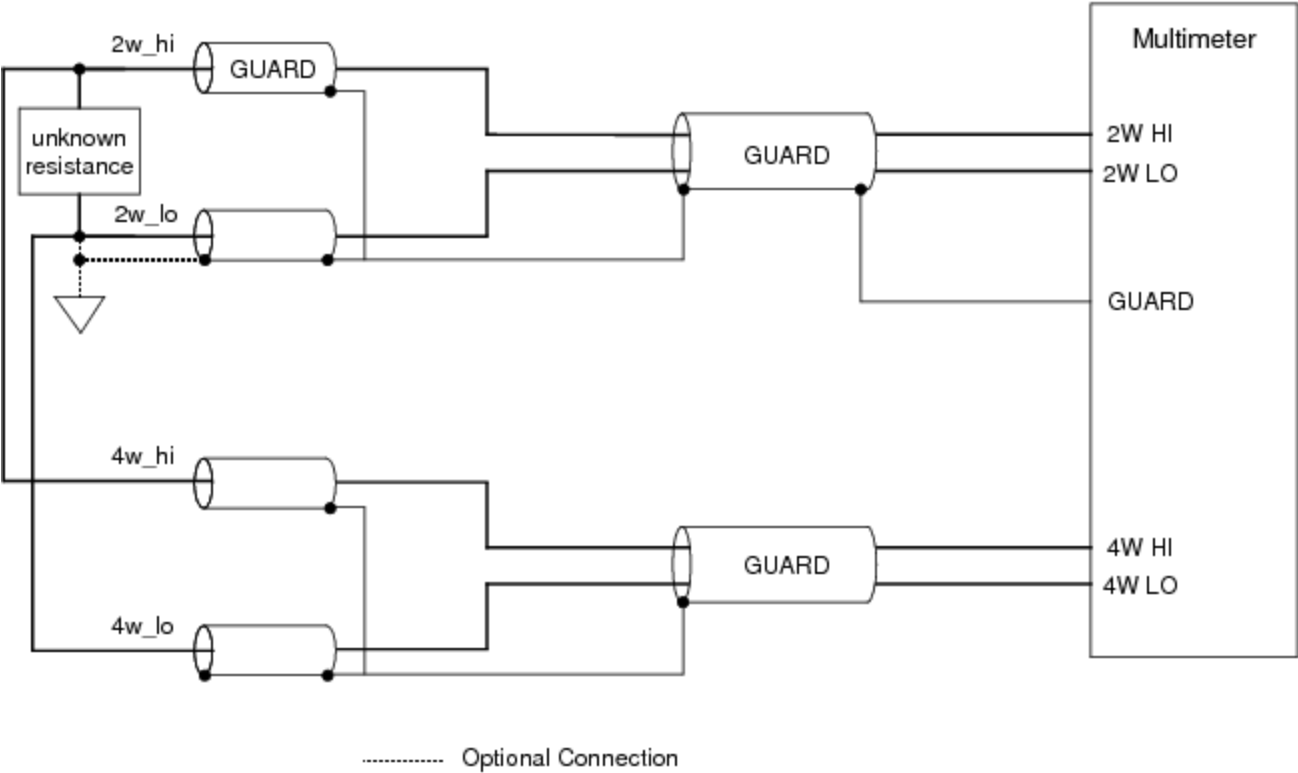
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PMO Theory of Operation : [Types of Measurement](#) : Four-Wire Resistance Measurements

Four-Wire Resistance Measurements

For resistances below 1 Mohm, the four-wire configuration significantly reduces the loss in measurement accuracy due to the loop resistance between the DUT and meter. This figure shows the four-wire configuration for resistance measurements.



Four-Wire Resistance Measurement

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